Computer Science - Undergraduate Level Assessment 5 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. What is the output of 4*1**4 in python?
   (a) 4
   (b) 1
   (c) 256
   (d) 16

2. Which of the following sets of bit strings CANNOT be described with a regular expression?
   (a) All bit strings whose number of zeros is a multiple of five
   (b) All bit strings starting with a zero and ending with a one
   (c) All bit strings with an even number of zeros
   (d) All bit strings with more ones than zeros

3. In python 3, which of the following is floor division?
   (a) /
   (b) //
   (c) %
   (d) |

4. Let l = [1,2,3,4]. What is max(l) in Python 3?
   (a) 1
   (b) 2
   (c) 3
   (d) 4

5. Let f(X) = if x = 1 then 0 else [x * f(x - 1) + x**2]. The value of f(4) is
   (a) 53
   (b) 29
   (c) 50
   (d) 100

True / False (10 marks)

Answer True or False

6. True or False: In a complete K-ary tree of depth N, the ratio of nonterminal nodes to total nodes is approximately K-1/K.
7. True or False: O(log n) indicates a smaller time complexity than O(1) in asymptotic analysis.
8. True or False: A 3-way, set-associative cache is effective only if 3 or fewer processes are running alternately on the processor.
9. True or False: The function O(log N) grows slower than O(log log N), making it a better choice for optimizing performance.
10. True or False: The maximum possible degree of the minimal-degree interpolating polynomial p(x) for n + 1 distinct points is n.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to search a literals string in a string and also find the location within the original string where the pattern occurs.
12. Write a python function to count the number of digits in factorial of a given number.

End of Paper